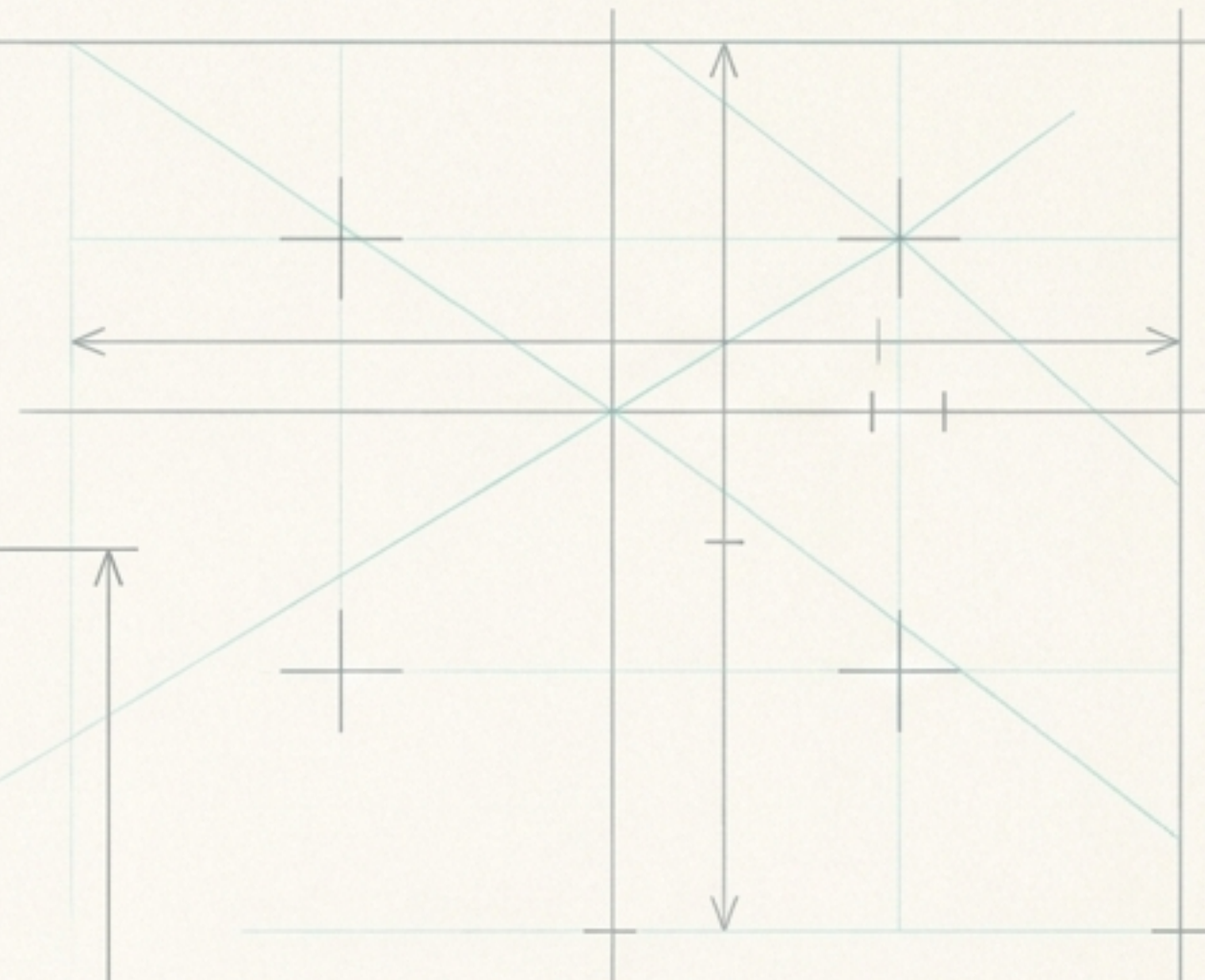
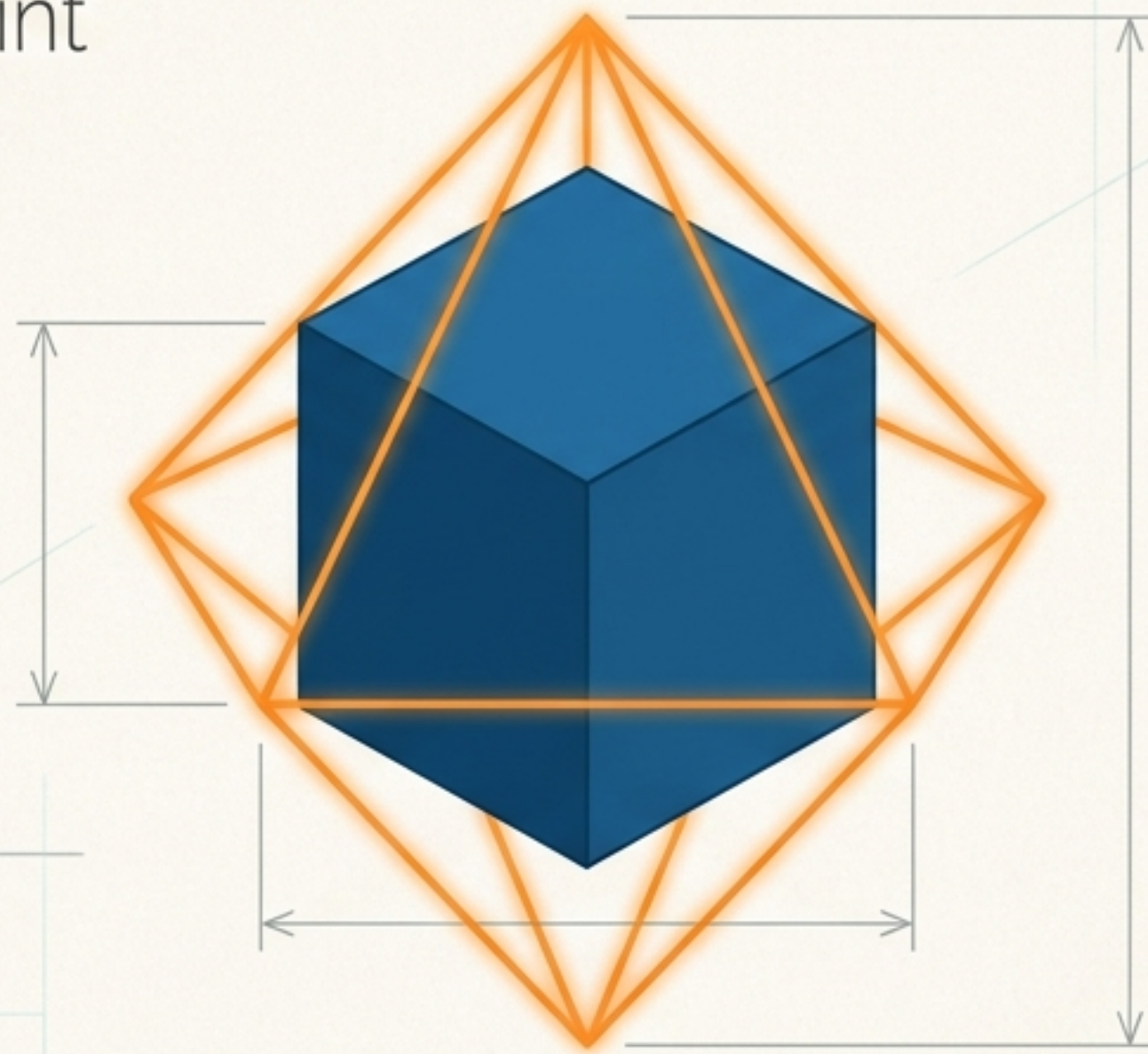


# Cracking the Symmetric Mahler Conjecture

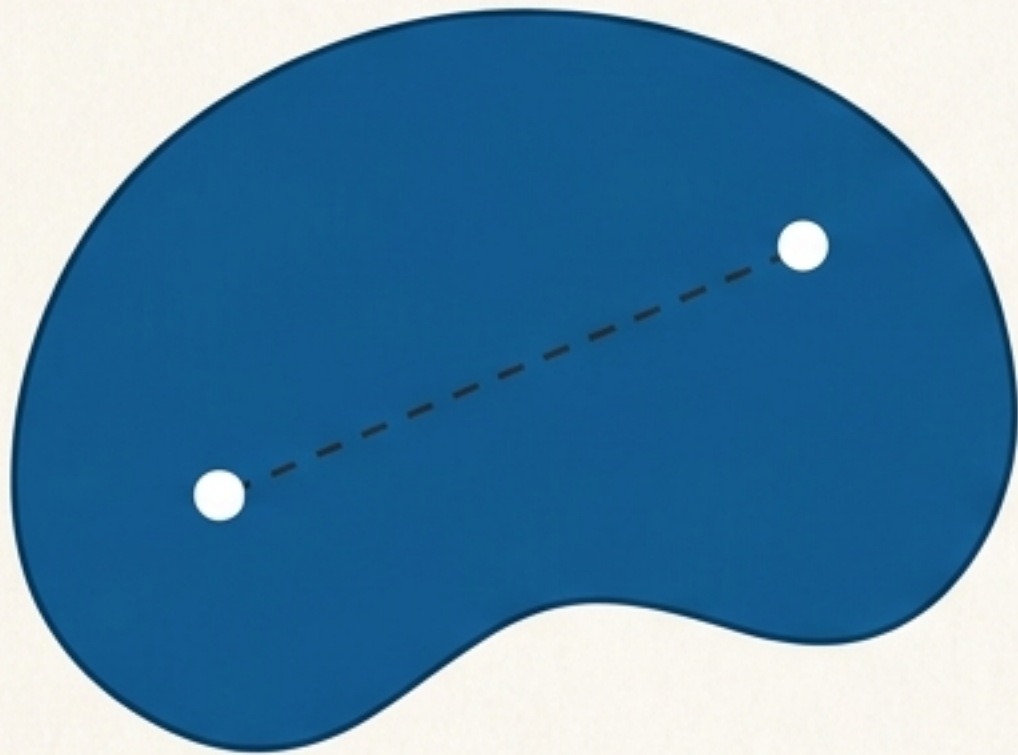
The Geometric Blueprint  
and the Analytic Lens



Based on the September 22, 2026,  
AI-generated OpenAI preprints  
(Open before this paper;  
currently not peer-reviewed).

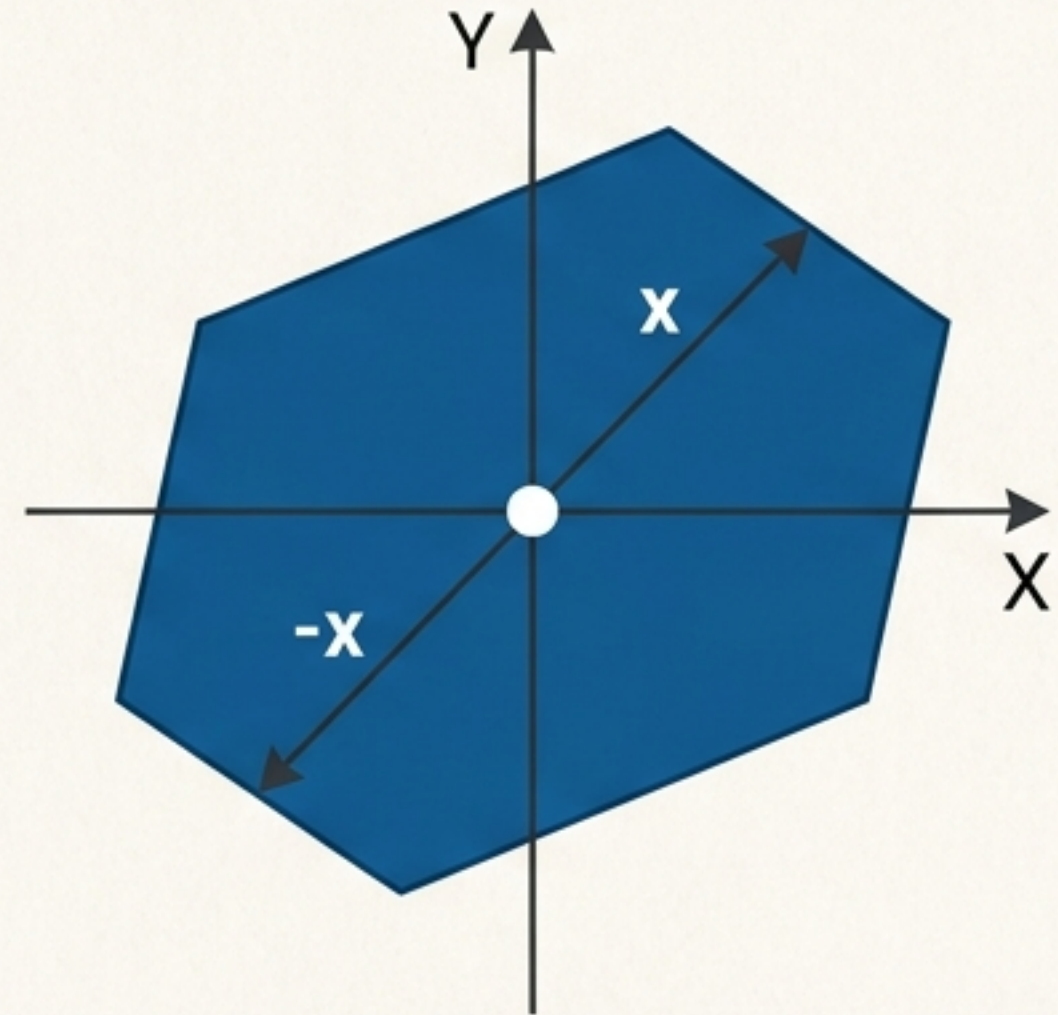
# The Rules of the Space: Convexity and Origin Symmetry

## 1. Convexity



**Convex:** Any line segment connecting two points inside the body remains entirely inside.

## 2. Origin Symmetry



**Origin-Symmetric:** For every vector  $x$  inside the body, its exact opposite  $-x$  is also inside.

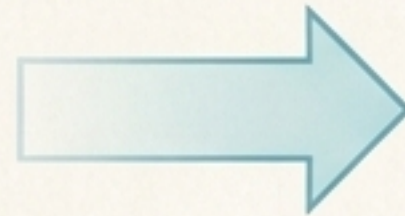
# The Dual World: The Polar Body $K^\circ$

$$K^\circ = \{y \mid x \cdot y \leq 1\}$$

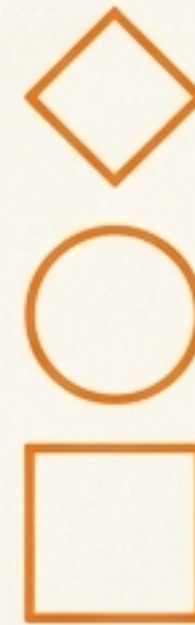
The polar body  $K^\circ$  is the set of all vectors  $y$  whose dot product with every vector  $x$  in  $K$  is  $\leq 1$ .

## Translation Matrix

Primal Space ( $K$ )



Polar Space ( $K^\circ$ )



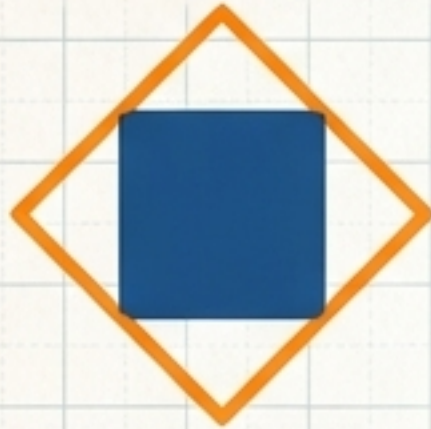
Sharp corners in the primal world become flat edges in the polar world, and vice versa.

# The Volume Product: A Dimensionless Invariant

**The Metric:**  $|K| |K^\circ|$  (Volume of  $K \times$  Volume of  $K^\circ$ )

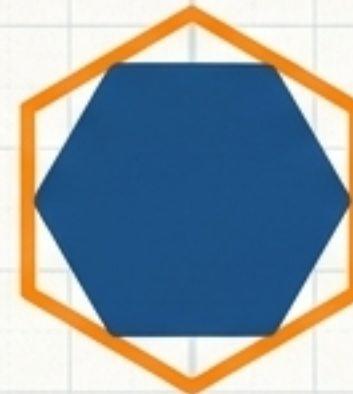
**The Core Rule:** This product is entirely unchanged by invertible linear transformations (stretching, skewing, rotating). A skewed square has the exact same volume product as a perfect square.

Minimizers  
(Conjectured)



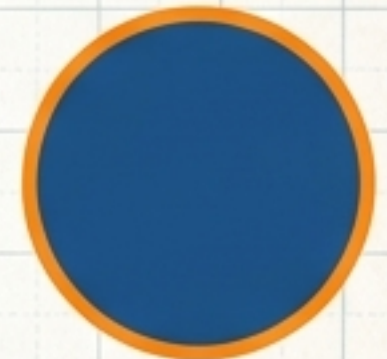
8

Intermediate



9

Maximizer  
(Solved)



$\pi^2 \approx 9.87$

# The Extremes of the Universe

## Blaschke-Santaló Inequality



The maximum volume product is always achieved by balls and solid ellipsoids.

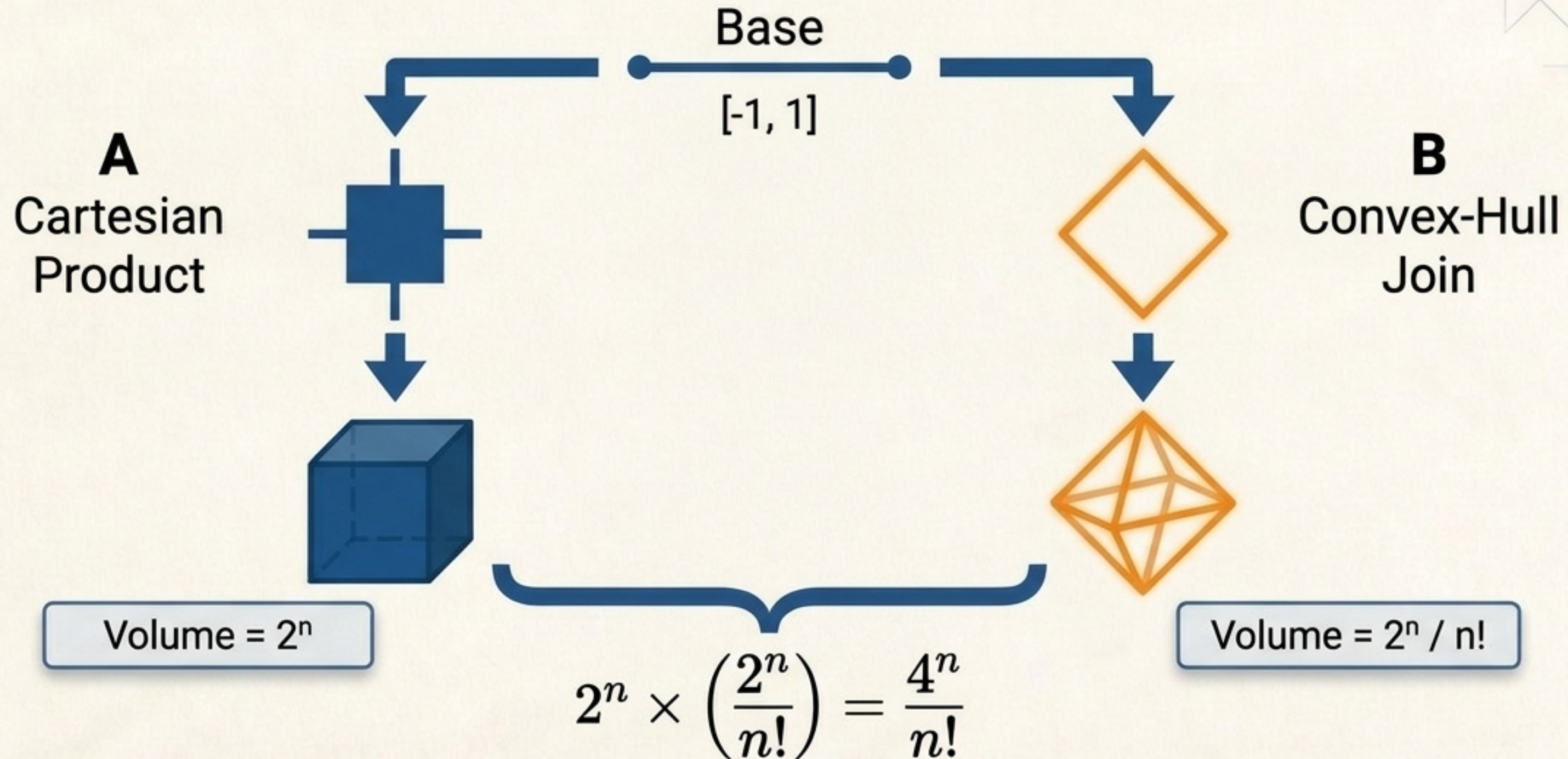
## Mahler's Conjecture



The minimum volume product is conjectured to be exactly  $4^n / n!$

While the upper bound has essentially one extremizer (the ball), the lower bound contains a vast, chaotic family of suspected minimizers...

# Meet the Suspects: Hanner Polytopes



Hanner polytopes are built recursively from 1D segments using products and joins. Mahler conjectured that all of these achieve the absolute minimum.

# A 90-Year Timeline of the Symmetric Conjecture

● **1938-1939:** Kurt Mahler proves the 2D planar case.

● **1987:** Bourgain & Milman establish a reverse Santaló inequality ( $c^n$ ), proving the volume product scales like a hypercube, but without the sharp constant.

● **2008:** Kuperberg explicitly improves the absolute constant.

● **2020:** Iriyeh & Shibata prove the exact 3-dimensional case.

““ An open problem for nearly a century, cracked by an AI-generated proof. ””

**Sept 22, 2026:** OpenAI preprints drop, claiming the proof for all dimensions.

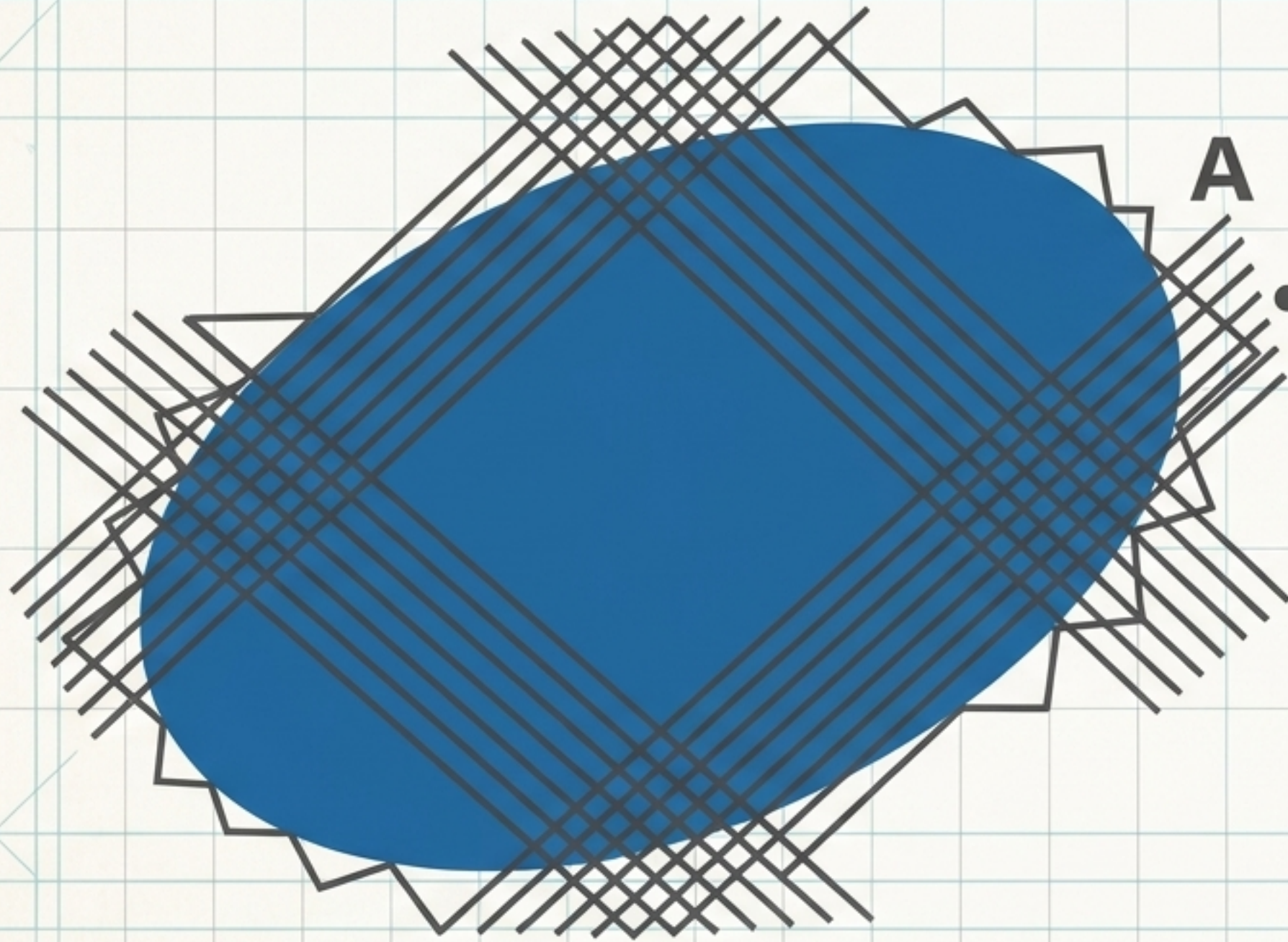
# The Main Theorem (2026)

For every integer  $n \geq 1$  and every origin-symmetric convex body  $K \subset \mathbb{R}^n$ :

$$|K||K^\circ| \geq \frac{4^n}{n!}$$

Equality holds if and only if  $K$  is a linear Hanner body (an invertible linear image of a Hanner polytope).

# Proof Step 1: The Slab Polytope

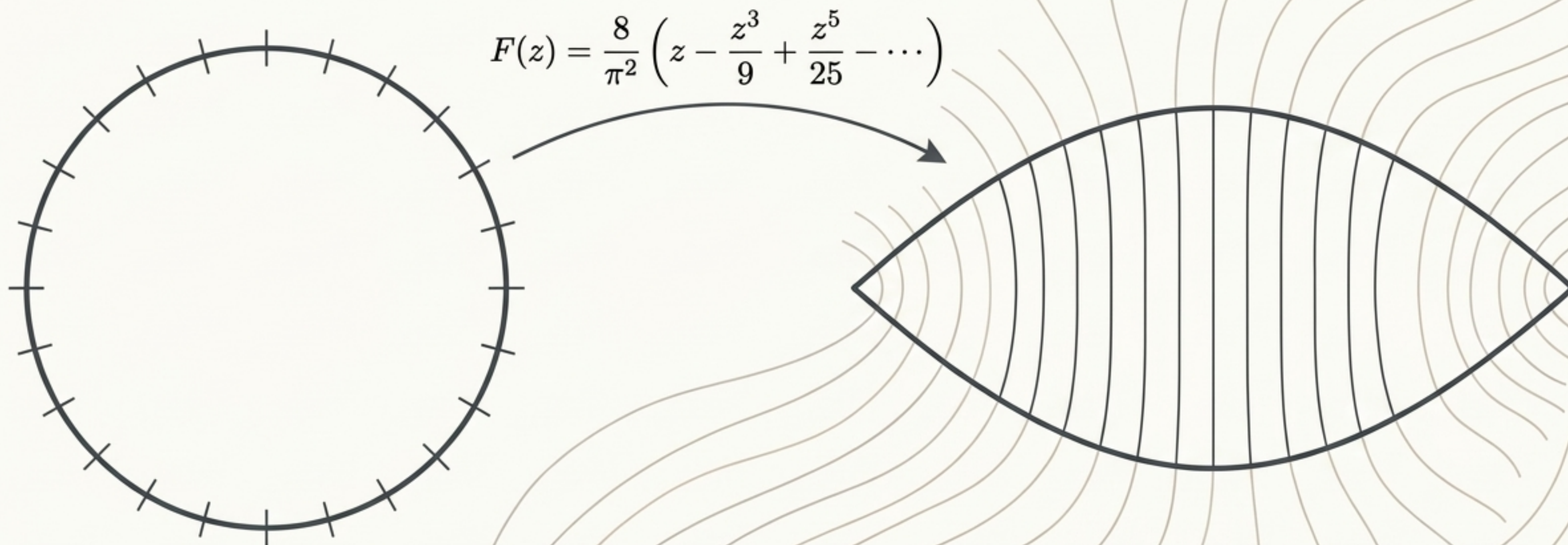


The proof begins by approximating the primal body  $K$  with a slab polytope:

$$A = \{|b_i \cdot X| \leq 1\}$$

This reduces the continuous volume to a discrete set of constraints, setting the stage for probability sampling.

# Proof Step 2: The Conformal Lens



1. The analytical breakthrough:  
A complex biholomorphic map  $F(z)$ .

2. This "Lens" sends uniform angular  
measures on the boundary to  
uniform imaginary heights.

3. By mapping discrete geometric parameters  
into the complex plane, probabilities ( $P_I$ ) of  
feasible regions can be precisely summed.

# Proof Step 3: The Crucial Bounds Collision

## The Geometric UPPER Bound.

Writing  $Z = Y + iX$ , each success forms a non-overlapping simplex  $\text{conv}(0, \pm b_i)$  inside the polar. Their total missing volume dictates the maximum bound.

$$S \leq \frac{n!}{4^n} |A| |A^\circ|$$



$S$

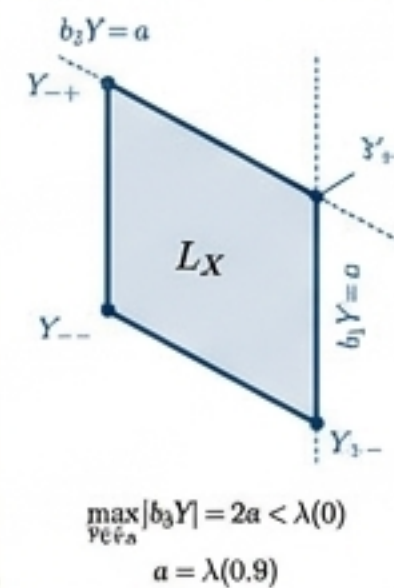


## The Analytic LOWER Bound.

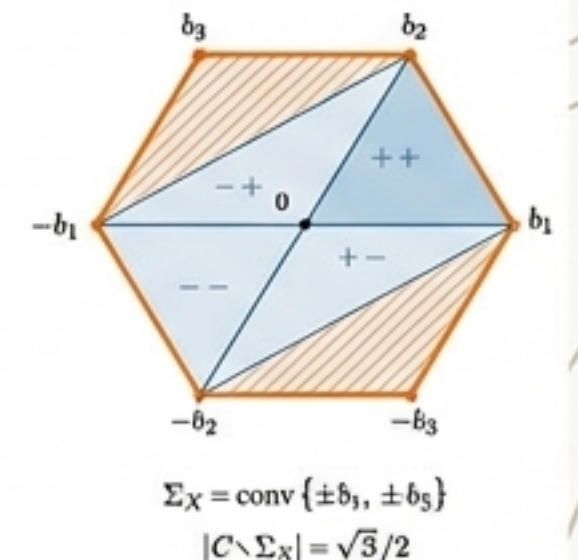
Derived via the holomorphic mass lemma on high powers of the inverse map.

$$S \geq 1$$

(a) Feasible witnesses in  $L_X$



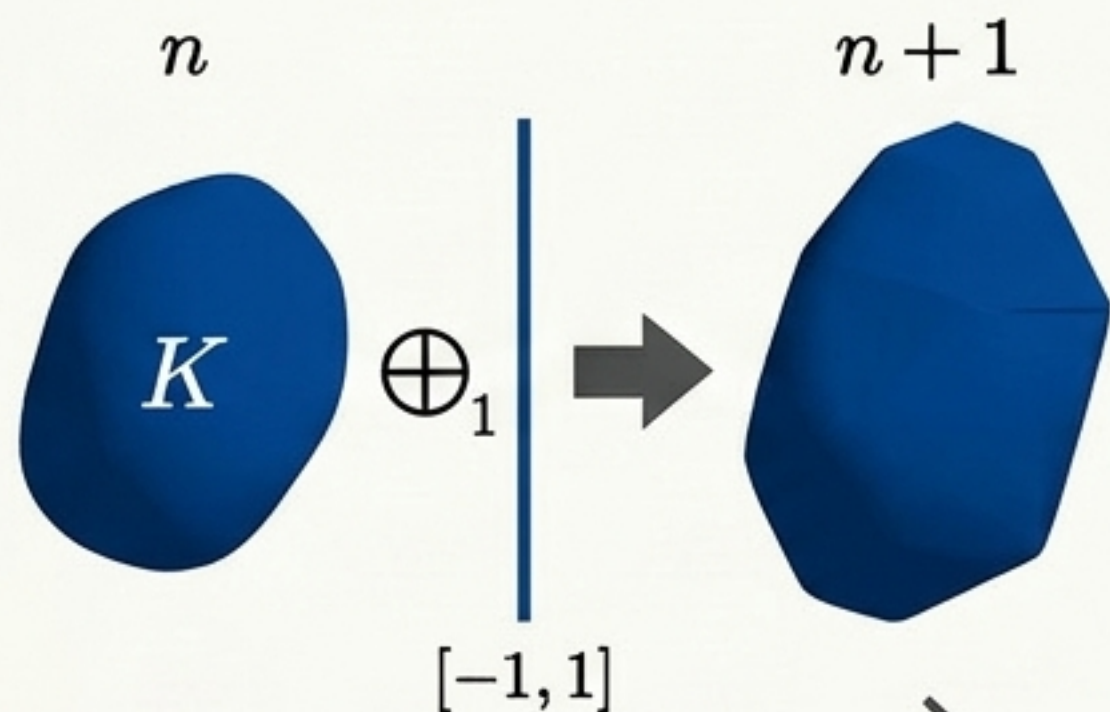
(b) Feasible simplices in  $C$



The rigid collision of the complex analytic lower bound and the discrete geometric upper bound proves the inequality. CRITICAL: Never swap these bounds.

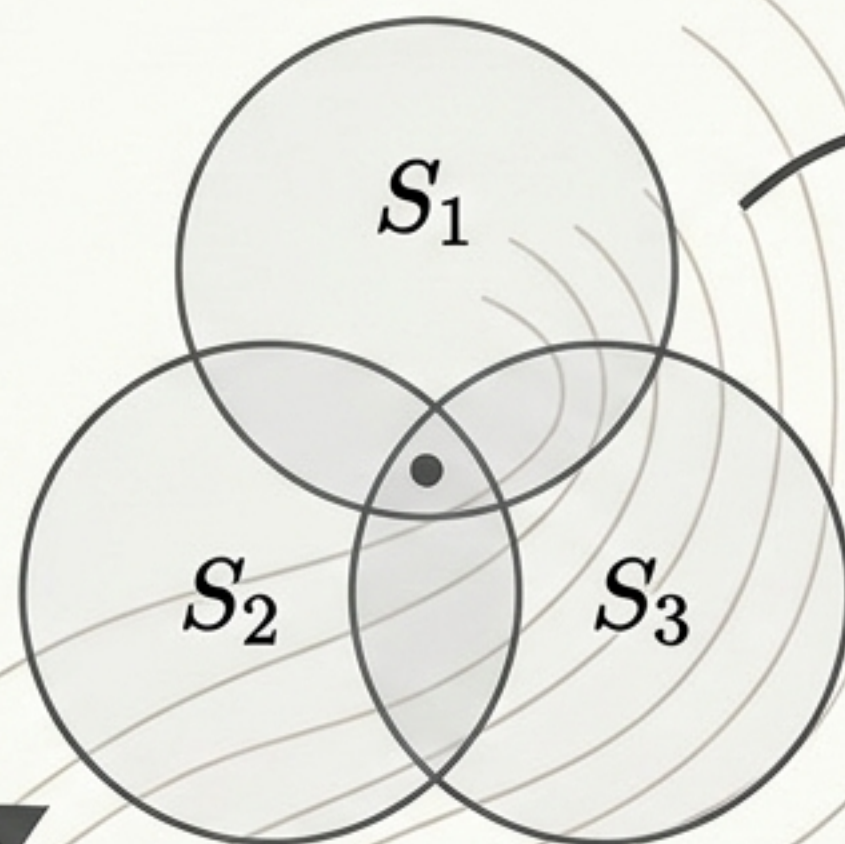
# Proof Step 4: The Equality Case

## The Lift



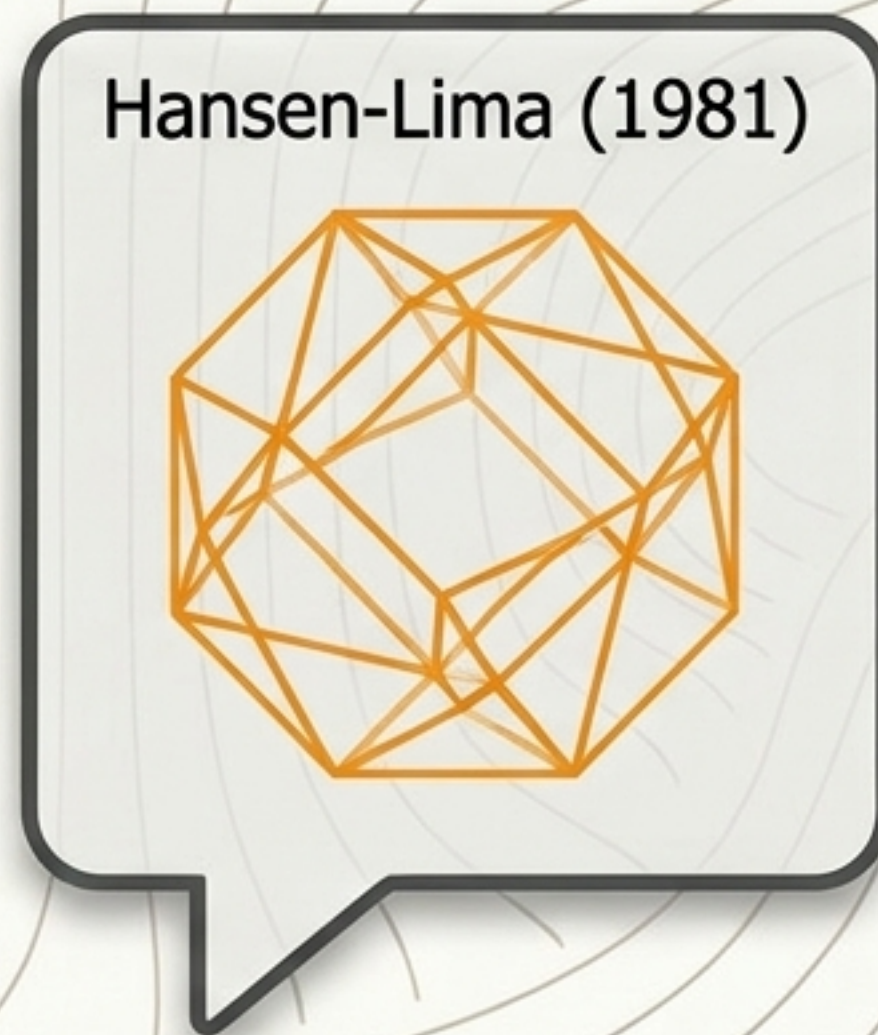
The base body  $K$  is lifted into dimension  $n+1$ , forcing the missing volume to vanish.

## The Metric



The geometry forces a unique metric median structure: the "Three-ball property".

## The Identification



Classical Banach space theory finishes the classification: exact minimizers must be invertible linear images of Hanner polytopes.

# The Architecture in Retrospect: Scope & Companions

A clean, direct proof bypassing a century of expected techniques.

## What the Proof DOES NOT Use

- ~~Steiner Symmetrization (belongs to Santaló's maximum proof, not the minimum).~~
- ~~Shadow flows.~~
- ~~Induction on dimension.~~

## Companion Papers (Separate AI Proofs)

Block 1 (The Non-Symmetric Case)  
Minimal volume product  
achieved by simplices, with  
constant  $(n + 1)^{n+1} / (n!)^2$ .



Block 2 (Symplectic Geometry)  
The Gromov width of  $K \times K^\circ$  is  
exactly 4.

